

Code Description

Python Files present:

- Order_model.py
- Order.py

Order_model.py:

1. get_features(): (Code is fully implemented/given)
 - a. Parameters:
 - i. filepath: path of the audio file
 - ii. sr (samplingrate = 8000) : for all the recordings given and newly recorded audio files
 - iii. using the program are recorded with a sampling rate of 8000
 - iv. n mfcc: 30
 - v. n mels: 128
 - vi. frames: 15
 - b. Details: Returns features, given the file path of the audio
2. Net: (Define your architecture)
 - a. Parameters : NA
 - b. Details: This Class shall be used to define your CNN architecture
3. record_voice() : (Code is fully implemented/given)
 - a. Parameters:
 - i. Username: an id given to a person
 - ii. j: label of the sample
 - iii. v: number of the sample recorded by a person with a particular ID
 - b. Details: This function when invoked records the audio from the system's default microphone for a second
4. play_audio(): (Code is fully implemented/given)
 - a. Parameters:
 - i. Path: path of the audio file to be played
 - b. Details: This function plays an audio (.wav) file given its path

Order.py: (Code partially implemented/ training classifier has to be coded by you)

1. Class order():
 - a. Methods:
 - i. classify_input(): (You have to Complete this function so that it returns predicted label)
 1. Parameters: Define your own parameters, as per your classifier model
 2. Details: While ordering the product, this method has to classify the input voice sample

- ii. Confirm input(): (Code is fully implemented/given)
 - 1. Parameters:
 - a. digit: predicted label for the given voice sample
 - b. flag : to track the menu and quantity
- iii. take user input(): (Code is fully implemented/given)
 - 1. Parameters: flag
 - 2. Details:
 - a. It prompts for the menu/quantity, records the value using record voice function in Order_model.py and extracts features from get_features () function in Order_model.py files
 - b. By passing these extracted features to classify_input() method, you have to predict the label.
 - c. Then the predicted digit is passed to confirm_input() function to check whether the predicted label is valid and returns the menu choice and label, otherwise calls the take_user_input() again to start the whole process once more.